

Seed Unfold

Technical project documentation for a
traceable story-world development engine



Seed Unfold

Story-world development and brainstorming/prototyping engine

This document defines the finalized hackathon product concept, user experience, MVP architecture, implementation plan, demo flow, and path to production.

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2. Executive Summary

Seed Unfold turns one seed idea into a choice-driven, traceable story-world exploration. A user enters one premise, fragment, question, or image prompt; the system interprets it and generates exactly three distinct possible worlds. The user compares those directions, selects one, and watches the selected branch unfold into a coherent mini-universe containing story scenes, characters, visual concepts, optional sound, and optional video prototypes.

The product is a story-world development and brainstorming/prototyping engine—not merely a one-click story generator and not necessarily a full movie generator. Its value is the structured movement from ambiguity to direction: possibility is expanded first, human judgment chooses the path, and downstream content inherits a visible creative source of truth.

Long-form stories can expand chapter by chapter while preserving characters, rules, tone, geography, timeline, unresolved threads, and provenance. Every generated element remains connected to the original seed, the selected world, relevant canon facts, and the user decisions that shaped it.

Product thesis

Seed Unfold makes creative possibility visible, selectable, coherent, and traceable.

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3. Project Concept and Core Idea

Core loop: one seed → three possible worlds → human choice → one coherent universe → inspectable creative evolution.

The seed is intentionally small and incomplete. Instead of forcing an immediate answer, Seed Unfold exposes several meaningful interpretations before commitment. Each candidate world has its own premise, rules, tensions, visual identity, risks, and opportunities, allowing the creator to compare directions rather than accept the first plausible completion.

The design principle is simple: AI expands possibility, while the human selects direction and retains authorship. Branching before commitment protects ambiguity as a creative asset. Once a direction is chosen, the system spends generation effort on coherence, not on endlessly producing disconnected alternatives.

Explore

Expand the seed into distinct interpretations.



Choose

Unfold

Develop one direction with continuity.

4. Problem Statement

Early creative ideas are ambiguous by nature, but most generative tools collapse that ambiguity too quickly into one answer. One-shot generation can produce attractive prose while allowing tone, characters, rules, and causality to drift. Creators often cannot see why a detail appeared, which source constraint it inherited, or how to revise it without breaking everything downstream.

Long-form expansion magnifies the problem: names change, relationships reset, world rules contradict earlier scenes, and promising branches disappear because the system has no durable memory of the decisions that mattered. Prototypes also jump too quickly to expensive media generation before the underlying direction has been tested.

The opportunity is to make exploration structured, visual, explainable, persistent, and economical. Seed Unfold treats text structure and provenance as the foundation, then adds images, audio, and video as optional layers once a human-approved direction exists.

Opportunity

Preserve ambiguity early, make choice explicit, and carry approved creative logic through every later artifact.

5. Tattva 2 — “Forms Hidden in Formless”

The connection is used as a product metaphor, not as a religious claim. A seed is formless potential: it contains direction without yet specifying a complete world. The understanding pass extracts latent themes, tensions, entities, settings, and constraints. Three world candidates then reveal distinct forms hidden inside the same input.

Human selection gives one form direction. The unfolding stage turns that selected possibility into a living, coherent system of rules, characters, scenes, and media references. Seed DNA and traceability preserve continuity so the form remains connected to its origin as it grows.

Tattva-inspired idea	Product behavior
Formless potential	Seed input
Hidden forms	Three world candidates
Discernment	Human selection
Unfolding	Structured generation
Continuity	Seed DNA and traceability

6. Product Definition and Actual Use

Seed Unfold is an interactive creative development workspace for writers, filmmakers, game designers, educators, facilitators, and hackathon teams. It combines structured ideation, world-building, narrative planning, optional media prototyping, and inspectable provenance in one calm workspace.

In actual use, a creator enters a rough premise, inspects three radically different interpretations, chooses one, develops its world bible, generates an initial story arc and scenes, creates supporting characters and media prototypes, inspects provenance, and then refines or branches without losing previous work.

The workspace is designed for people who have ideas but struggle to structure them. It does not ask users to become prompt engineers. It exposes the important decisions as visible controls: tone, rules, constraints, canon facts, branch points, and the reason an output exists.

- **Primary value:** move from vague idea to a defensible creative direction.
- **Working style:** text-first, choice-led, media-optional, history-preserving.

7. Exact End Users and Use Cases

User	Need	Example use
Writer	Turn an idea into a durable story system.	Develop a novel premise, character arcs, and chapter plan.
Filmmaker	Test directions before production.	Compare worlds, visual identity, and proof-of-concept scenes.
Game designer	Explore lore, quests, and rules.	Generate a world bible and branching quest seed.
Educator	Teach narrative and world-building.	Show how one premise becomes several coherent interpretations.
Creative team	Compare directions together.	Review options before committing budget or production time.
Facilitator	Run structured ideation.	Use three candidate worlds as workshop discussion anchors.
Solo creator	Structure an idea without a large team.	Prototype a pitch with traceable creative logic.

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8. Ultimate User Output

The output is a chosen, coherent mini-universe package: world premise, rules, themes, characters, relationships, chapter and scene plan, selected generated visuals, optional audio or video prototypes, and a traceability graph. It is an organized creative workspace, not a transcript of prompts and responses.

Seed	Three worlds	Selected branch
A child discovers a forgotten city beneath the ocean.	Lost Civilization Bio-City Time Capsule	Child protagonist, archivist guardian, submerged city rules, scene outline, visual references, optional atmospheric sound.

- **World logic:** the selected branch defines what is possible and what is unusual.
- **Story logic:** scenes inherit the selected world, approved characters, and active constraints.
- **Traceability:** every generated element points back to the selected world and ultimately to the seed.

9. Core USP and Differentiators

Dimension	Seed Unfold	Common alternative
Direction	Exactly three world directions before commitment.	One immediate completion.
Agency	Human choice is a first-class product step.	Choice is hidden in a prompt or rewrite.
Consistency	Seed DNA, canon facts, and branch history.	Context is mostly implicit.
Explainability	Traceability and “Why was this generated?”	Opaque output provenance.
Expansion	Chapter-by-chapter growth with continuity checks.	Disconnected generations.
Media	Optional multimodal layers after text structure.	Media generation without a stable world.
Scope	Prototype-first, not a full-movie promise.	Overclaiming end-to-end production.

Why it matters

The differentiator is not only generation quality. It is the combination of branching, human authorship, continuity, persistence, and an answerable reason for every important output.

10. MVP Scope vs Future Features

MVP	Future / non-MVP
Text seed input; seed understanding; exactly three world candidates; world selection; structured Seed DNA; core world bible; characters and relationships; short story and scene generation; simple visuals via image API; traceability graph; refine and branch; save/load persistence; deterministic fallback/demo mode; basic responsive UI; one polished end-to-end demo path.	High-fidelity video generation; full-length film production; real-time collaboration; advanced memory across huge books; voice cloning; marketplace; production asset management; complex animation; autonomous multi-agent orchestration; publishing and distribution integrations.

MVP boundary

TinyStories is not the primary LLM. It is a simple children’s-story dataset/model approach useful for experiments or educational demos, not the main choice for complex world-building. The primary LLM should be a capable general model such as Gemini, with separate generation APIs for media.

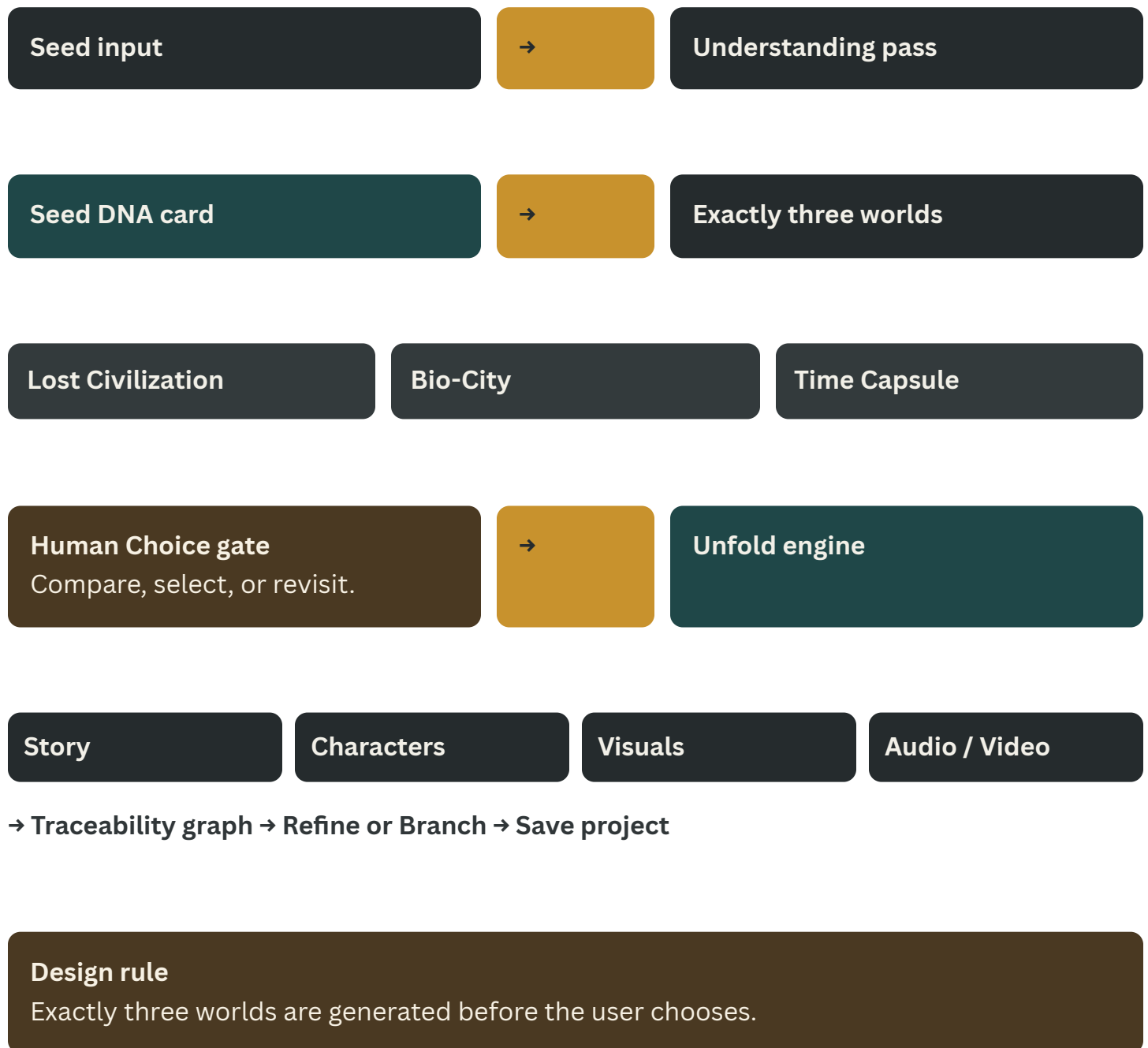
11. Complete End-to-End Workflow

Overview: Seed → Understand → Seed DNA → 3 Worlds → Human Choice → Unfold → Story / Characters / Visuals / Audio / Video → Traceability → Refine / Branch → Save

1. **Seed** — The user enters a premise, fragment, question, image prompt, or rough idea. The system preserves the original wording as the creative source.
2. **Understand** — Extract themes, genre signals, tone, entities, conflicts, constraints, ambiguities, and opportunities into structured fields.
3. **Seed DNA** — Normalize the canonical creative source of truth: premise, themes, world rules, tone, style, key entities, open questions, and provenance.
4. **3 Worlds** — Generate exactly three meaningfully distinct directions with names, loglines, differentiators, rules, risks, and visual identity.
5. **Human Choice** — The user compares and selects one branch. Selection is explicit, reversible, and recorded as a decision.
6. **Unfold** — Expand the chosen world into a coherent mini-universe with a world bible, characters, relationships, and story structure.
7. **Story / Characters / Visuals / Audio / Video** — Generate structured text first, then add media as optional prototypes connected to the relevant creative element.

8. **Traceability** — Connect every artifact to the source seed, selected world, parent scene, rule, or user decision.
9. **Refine / Branch** — Edit constraints, regenerate a creative element, or branch without destroying the original version.
10. **Save** — Persist the project, branch history, Seed DNA, generated assets, graph, and job statuses.

12. Clear Workflow Diagram



→ Traceability graph → Refine or Branch → Save project

13. User Experience and Clean Final-Output UI Concept

The application uses a calm, dark, content-first workspace. A left rail handles project and branch navigation; the main canvas shows the current world, story, or scene; a right inspector exposes Seed DNA, constraints, provenance, and “Why was this generated?”; a bottom action bar provides refine, branch, save, and export actions.

Screen	Primary job
Seed entry	Capture the original idea and optional context.
Three-world comparison	Compare names, loglines, rules, risks, and visual identities.
Selected-world dashboard	Show the world bible, canon, characters, and next actions.
Scene detail	Inspect prerequisites, consequences, assets, and generated text.
Traceability inspector	Answer why an item exists and which decisions support it.
Saved branch history	Reopen prior states, compare branches, and restore a version.

The final output should feel like an organized creative workspace rather than a chatbot transcript. Progressive disclosure keeps the first view focused while allowing a curious user to open rules, source context, graph edges, prompts, or generation status when needed.

14. Seed DNA Design and Consistency Mechanism

Seed DNA is a structured, versioned object that travels through the pipeline. It is not just a prompt; it is the consistency and explainability layer that carries approved meaning into every generation step.

Field	Purpose
seed_id	Stable identity for the original creative source.
canonical_seed	Preserved user wording and normalized premise.
extracted_themes, genre, tone, audience	Creative signals used for controlled generation.
world_rules, constraints, visual_style, sound_style	Boundaries and sensory direction.
key_entities, relationships	People, places, objects, and their links.
unresolved_questions, selected_world_id	Open design space and chosen direction.
canon_facts, branch_id, provenance	Approved continuity, version context, and source links.

Every generation receives the relevant Seed DNA context. Canon facts can be locked or user-approved; contradictions are flagged rather than silently overwritten; versions allow safe refinement and branch creation. A scene can therefore be regenerated without losing the source rules that made it belong to the selected world.

15. System Architecture

Client layer

React + Vite + Tailwind + Framer Motion

↓ synchronous text generation and user decisions

API / orchestration layer

FastAPI routes · authentication/session · project service · branch service · generation orchestrator · traceability service

↓ structured context and provider calls

General LLM

Understanding,
branching, story

Image API

Concept visuals

TTS / audio

Narration,
ambience

Validators

Schemas, safety,
fallback

↓ persistence, asynchronous media jobs, provenance events

Data layer

Relational database for projects, worlds, scenes, entities, graph edges, decisions, and versions.

Object storage for images, audio, video, and exports.

Cache and job queue for asynchronous generation.

Deployment layer

Web hosting/CDN, containerized FastAPI service, managed database/storage, secrets management, logging, metrics, and rate limits.

16. AI Pipeline and Model Roles

1. Parse and classify the seed into structured signals.
2. Generate a structured understanding and Seed DNA draft.
3. Create three diverse world candidates, then validate distinctness and schema.
4. Wait for explicit human selection before expensive unfolding.
5. Build the selected world bible, plan scenes, and generate characters.
6. Generate visual prompts and submit optional media jobs.
7. Run consistency checks, write traceability edges, and persist outputs.

Model roles

- **Gemini or another capable general LLM** — understanding, branching, story planning, structured JSON, and controlled rewriting.
- **Image generation API** — concept frames, character references, and world visuals.
- **TTS/audio generation API** — narration, ambience, and short atmospheric layers.
- **Optional video API** — short proof-of-concept clips, not a core MVP dependency.
- **Deterministic templates and fixtures** — reliable fallback and demo mode.

TinyStories is a simple children's-story dataset/model approach useful for experiments or educational demos. It is not the primary model for complex world-building.

17. Recommended Technology Stack

Layer	Recommended technology	Role
Frontend	React, Vite, Tailwind CSS, Framer Motion	Responsive dark workspace, transitions, comparison cards, graph and UI states.
Backend	Python FastAPI	Typed APIs, orchestration, validation, and persistence integration.
LLM	Gemini or comparable capable general model	Understanding, world branching, story, scene and character generation, structured output.
Images	Image generation API	Concept frames, character references, and world visuals.
Audio	TTS/audio generation API	Narration and atmosphere.
Video	Optional video generation API	Short clips only, not a core MVP dependency.
Data	PostgreSQL or equivalent relational database	Projects, branches, Seed DNA, entities, scenes, decisions, and versions.
Storage	Object storage	Generated media and export assets.
Async	Background jobs and queue	Media generation, retries, and status updates.
Deployment	Containerized service plus managed database/storage, CDN, secrets, logs, metrics, and rate limits	Reliable delivery, operations, and controlled cost.

18. Data Model and Core API Contracts

Entity	Purpose
Project, Seed, SeedDNA	Container, original input, and canonical creative source.
WorldCandidate, WorldSelection	Three alternatives and the recorded human choice.
Character, Scene, Asset	Universe content and generated media references.
TraceEdge, Branch, UserDecision	Provenance, history, alternate paths, and authorship.
GenerationJob	Provider status, retries, outputs, and failure state.

Contract	Behavior
POST /projects	Create project with seed.
POST /projects/{id}/understand	Return structured understanding and Seed DNA draft.
POST /projects/{id}/worlds	Generate exactly three WorldCandidate records.
POST /projects/{id}/selection	Persist selected world and user rationale.
POST /projects/{id}/unfold	Generate world bible, characters, scenes, and media jobs.
GET /projects/{id}/trace	Return graph nodes and edges.
POST /projects/{id}/branches	Create branch from a selected node or version.
POST /generation-jobs/{id}/retry	Retry with fallback or provider switch.
GET /projects/{id}	Load current project, branch, canon, assets, and statuses.

Structured responses should be schema-validated, versioned, idempotent where practical, and linked to provenance IDs. Core relationships are explicit: a project owns seeds and branches; a selected world owns canon; scenes and characters reference the active branch; assets and jobs attach to source nodes; trace edges connect every decision and derivative.

19. Traceability Graph and “Why Was This Generated?”

The graph contains nodes for Seed, Understanding, Seed DNA field, World Candidate, User Choice, Canon Fact, Scene, Character, Asset, and User Edit. Edges include `derived_from`, `selected_by`, `constrained_by`, `appears_in`, `generated_for`, `revised_from`, and `branched_from`.

A child discovers a forgotten city beneath the ocean → extracted theme “memory and discovery” → selected world “Time Capsule” → canon rule “the city opens only during a tidal alignment” → Scene 01 “The child follows the blue lights” → concept image asset.

Inspector answer

“This scene was generated for the Time Capsule world because it inherits the seed’s discovery theme, the selected world’s tidal-alignment rule, and the approved protagonist.”

Users can inspect, revise, or remove the supporting constraint. The graph is therefore both an audit trail and a creative editing surface: provenance explains the current result, while edits create new traceable versions rather than erasing history.

20. Long-Form Expansion

Seed Unfold grows a universe chapter by chapter and scene by scene. Each chapter receives the current Seed DNA, canon facts, previous chapter summary, active character states, unresolved threads, and selected tone. The model is asked to extend a living system, not restart from a blank prompt.

Each scene stores prerequisites, consequences, entities present, visual and audio prompts, and trace edges. Continuity checks compare names, relationships, rules, timeline, geography, and unresolved questions before the scene is approved or added to canon.

Users can expand one chapter, regenerate one scene, or branch from a point while preserving the parent version. A branch may explore a different consequence, tone, or rule interpretation without corrupting the original story path.

Continuity input	Check
Characters and relationships	Names, roles, loyalties, injuries, and active goals remain coherent.
World rules and geography	Locations, constraints, and exceptions are not silently contradicted.
Timeline and threads	Events, prerequisites, consequences, and open questions remain ordered.

Seed Unfold — Technical Project Documentation

21. Reliability, Fallback, Persistence, Security, and Production Deployment

Reliability

- ☐ Schema validation for every structured response.
- ☐ Retries with exponential backoff, provider timeouts, and visible generation status.
- ☐ Distinctness checks for the three worlds, safety filters, moderation, and contradiction detection.
- ☐ Partial-save checkpoints after each major step.

Fallback and persistence

- ☐ Deterministic seeded fixtures for the ocean-city example, cached outputs, and local mock generation.
- ☐ Provider abstraction so the demo works when an external API is unavailable.
- ☐ Save after every major step; version Seed DNA; preserve branches, asset status, and resume state.

Security and production deployment

- ☐ Authentication, project-level authorization, encrypted secrets, signed media URLs, filtering, PII minimization, rate limits, and audit logs.
- ☐ Prompt-injection boundaries for imported content, containerized services, managed database and object storage, queue workers, CDN, observability, backups, migrations, cost controls, and feature flags.

22. Key Technical and Product Challenges with Mitigation

Challenge	Mitigation
Maintaining consistency	Seed DNA, canon locks, continuity validators, and versioned context.
Meaningfully distinct worlds	Diversity rubric, contrastive prompts, and post-generation comparison.
Explaining provenance	Graph edges, stable source IDs, and the inspector view.
Latency and cost	Staged generation, asynchronous jobs, caching, and a text-first MVP flow.
Unreliable providers	Retries, timeouts, provider abstraction, and fallback fixtures.
Media quality variance	Prompt templates, status visibility, and human selection.
User overwhelm	Progressive disclosure and a focused three-world comparison.
Scope creep	Explicit MVP boundaries and a reliable vertical slice.
Safety and ownership	Moderation, user controls, provenance, labeling, and policy review.

23. Social, Creative, and Educational Impact

Seed Unfold lowers the barrier between a vague idea and a structured creative direction. It supports divergent thinking before convergence, helping users see alternatives they may not have considered while keeping the final direction in human hands.

For learners, the product makes narrative causality, world-building, revision, and consequence visible. For teams, it creates a shared visual and textual artifact that can be discussed before production. For individual creators, traceability makes AI assistance more inspectable and less mysterious.

Showing three possible interpretations can also support alternative perspectives and encourage discussion about assumptions. The same mechanism introduces risks: homogenization, bias, over-reliance, authorship confusion, and media misuse.

Benefits

Lower entry barrier · divergent thinking ·
shared artifacts · narrative learning ·
inspectable assistance

Responsible controls

Human choice · transparent provenance ·
editable canon · generated-prototype
labels · user review

24. 24-Hour Hackathon Implementation Plan

Time	Focus
Hour 0–2	Finalize user journey, data shapes, ocean-city fixture, and visual direction.
Hour 2–5	Build React/Vite/Tailwind shell, routing, project state, seed entry, and three-world comparison.
Hour 5–8	Implement FastAPI endpoints, schema models, persistence, and deterministic fallback mode.
Hour 8–11	Integrate capable LLM for understanding, Seed DNA, exactly three worlds, and selected-world unfolding.
Hour 11–14	Add characters, scenes, traceability graph, and inspector.
Hour 14–17	Add image generation integration or polished mocked media, asset cards, and optional TTS/audio.
Hour 17–20	Add refine, branch, save flows, loading/error states, and consistency checks.
Hour 20–22	Polish UI, responsive A4-like documentation/export view, seed example, and demo script.
Hour 22–24	Test end-to-end, harden fallback, rehearse judge demo, capture screenshots, and finalize documentation.

Execution principle

Prioritize a reliable vertical slice over broad unfinished features.

25. Judge Demo Flow

1. **Enter the seed** — “A child discovers a forgotten city beneath the ocean.”
2. **Show understanding** — Reveal themes, genre signals, tensions, entities, and the first Seed DNA draft.
3. **Reveal exactly three worlds** — Lost Civilization, Bio-City, and Time Capsule.
4. **Compare directions** — Show how rules, tone, risks, and visual identity differ.
5. **Select one branch** — Choose a world and make the user decision visible.
6. **Unfold** — Generate the child protagonist, archivist guardian, submerged city rules, and first scenes.
7. **Add media** — Open a visual concept frame and optional atmospheric sound.
8. **Explain provenance** — Click “Why was this generated?” and reveal the supporting seed, world rule, and approved character.
9. **Refine and save** — Branch or refine one constraint, save, and reopen the project.

Closing thesis

Seed Unfold makes creative possibility visible, selectable, coherent, and traceable.

26. Definition of Done

- ☐ A user can create a project from one seed.
- ☐ The system returns exactly three distinct world candidates.
- ☐ The user can select one world.
- ☐ Seed DNA is visible and versioned.
- ☐ The selected world produces a coherent mini-universe.
- ☐ At least one story scene and character are generated.
- ☐ At least one visual or polished visual placeholder is available.
- ☐ Traceability explains a generated item.
- ☐ Refine and branch preserve history.
- ☐ Save and load work.
- ☐ Fallback and demo mode work without external providers.
- ☐ The architecture and API contracts are documented.
- ☐ The judge flow completes reliably.
- ☐ TinyStories is correctly positioned as non-primary.
- ☐ The result clearly communicates that Seed Unfold is not merely a one-click story generator or a guaranteed full movie generator.

27. Future Roadmap and Conclusion

Phase	Direction
Phase 1	Reliable MVP and text-first traceability.
Phase 2	Richer collaboration, more media controls, and improved continuity.
Phase 3	Long-form projects, stronger memory, evaluation, and export.
Phase 4	Ecosystem integrations, publishing workflows, and optional advanced video.

Seed Unfold begins with a single seed, exposes three possible forms, lets a human choose, and unfolds the chosen possibility into a coherent, inspectable universe. The product’s lasting value is not just generation; it is structured creative direction with continuity, agency, and traceability.